# Multiple-choice questions

|!|BCEMB0030000001|!| ★

A student observed some plant cells using a light microscope.

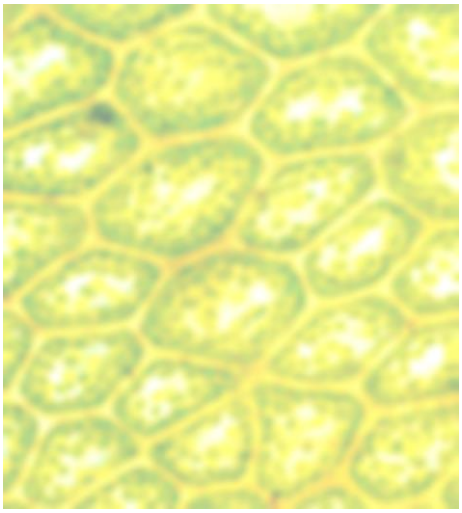


**Image A**

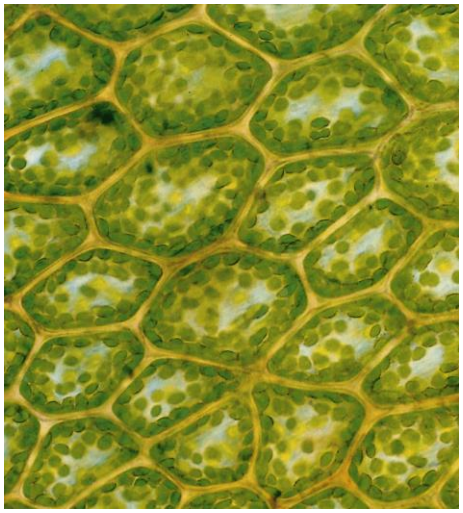


**Image B**

Initially, the student obtained an image as in A. In other to produce image B, which parts of the light microscope should be adjusted?

(1) objective
(2) fine adjustment knob
(3) iris diaphragm

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

| Skill assessed | Mark |
|---|---|
| Practical skills | 1 |

|!|BCEMA0030000002|!| ★★

A student studied a photomicrograph and an electron micrograph of a mesophyll cell. Which structure(s) can only be seen because of the higher resolution of the electron microscope?

(1) ribosomes
(2) the nuclear membranes
(3) grana in chloroplasts

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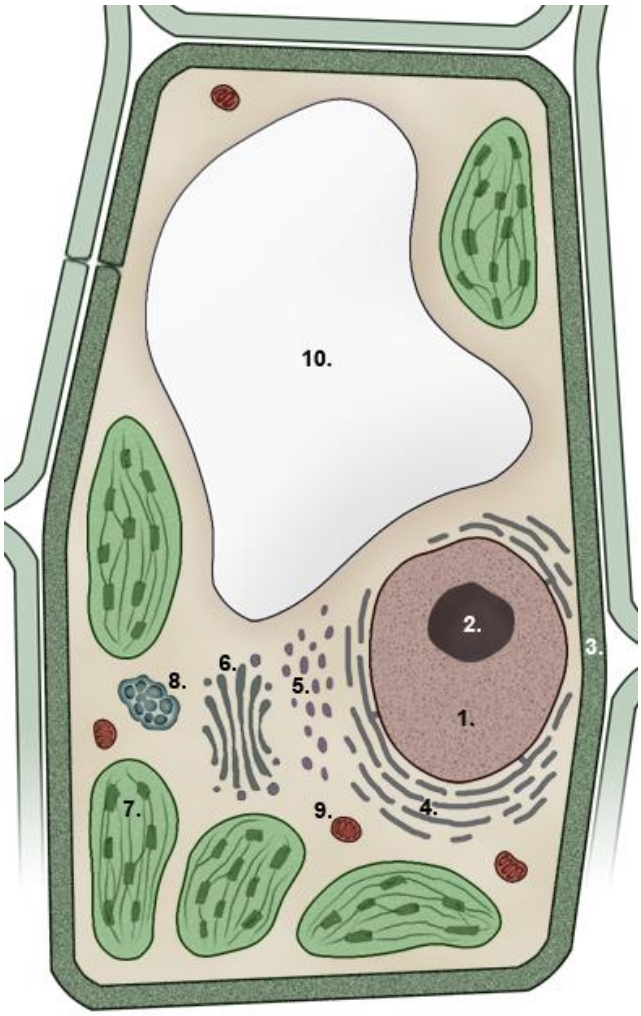
A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

##D##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMB0030000003|!| ★

The diagram below shows the structure of a plant cell viewed with an electron microscope.




(Credits: Vzb83, Wikimedia Commons, https://commons.wikimedia.org/wiki/File:Kasvisolu.png)

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What substances can be found in structures 2, 7 and 10?

| | *Structure 2* | *Structure 7* | *Structure 10* |
|---|---|---|---|
| A. | nucleic acids | chlorophyll | air |
| B. | nucleic acids | starch | cell sap |
| C. | starch | nucleic acids | cell sap |
| D. | glycogen | chlorophyll | water |

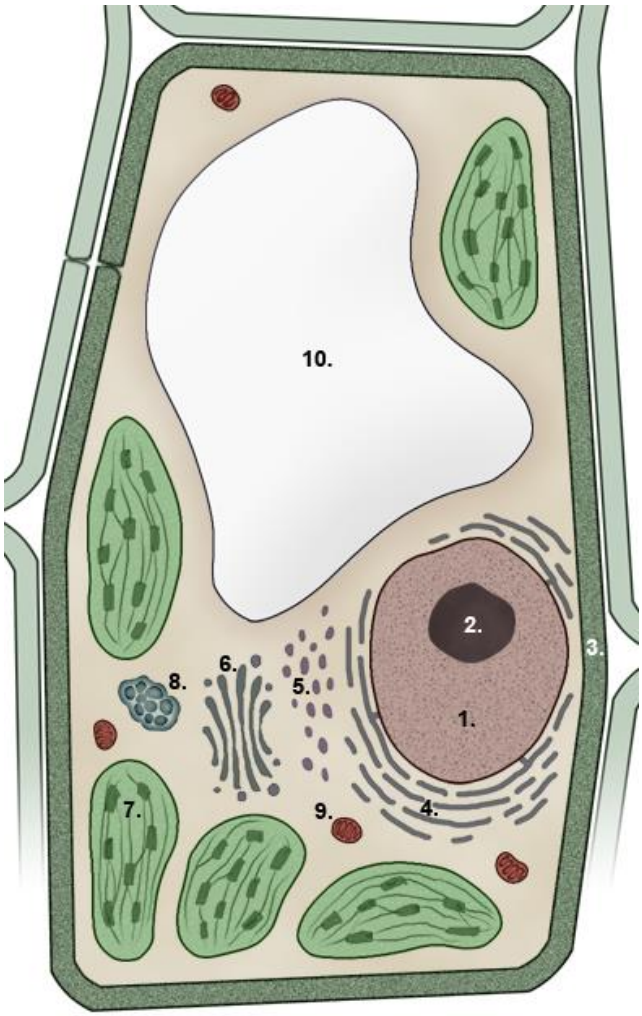
□

##B##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMB0030000004|!| ★

The diagram below shows the structure of a plant cell viewed with an electron microscope.




(Credits: Vzb83, Wikimedia Commons, https://commons.wikimedia.org/wiki/File:Kasvisolu.png)

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Which structures are absent in animal cells?

A. 3 and 10 only
B. 3, 7 and 9 only
C. 3, 7 and 10 only
D. 2, 3, 7 and 10 only

☐

##C##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMA0030000005|!| ★★

Which cell organelle does ***not*** contain nucleic acid?

A. chloroplast
B. cell membrane
C. mitochondrion
D. ribosome

☐

##B##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMC0030000006|!|

Some plant cells are stained and observed using a laboratory light microscope. Which of the following can be clearly visible at ×400 magnification?

A. mitochondria
B. chloroplasts
C. rough endoplasmic reticulum
D. ribosomes

☐

##B##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMB0030000007|!| ★

The diameter of the field of view of a light microscope is 2 mm. 20 plant cells were counted lining across the diameter. What is the average length of a plant cell?
(Hint: 1 mm = 1000 μm)

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A. 0.1 μm
B. 10 μm
C. 20 μm
D. 100 μm

##D##

| Skill assessed | Mark |
|---|---|
| Mathematical skills | 1 |

|!|BCEMC0030000008|!|

Which of the following correctly describes the plant cell walls?

A. They are composed of carbohydrates and are permeable to solutes.
B. They are composed of carbohydrates and are differentially permeable to solutes.
C. They are composed of phospholipids and are permeable to solutes.
D. They are composed of phospholipids and are differentially permeable to solutes.

##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000009|!|

Which of the following shows the correct order of organization in multicellular organisms with increasing complexity?

A. organ → tissue → system
B. organ → system → tissue
C. tissue → organ → system
D. tissue → system → organ

##C##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

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|!|BCEMB0030000010|!| ★

Which of the following observations supports the hypothesis that the nucleus is of great importance to cells?

A. Both animal cells and plant cells contain nuclei.
B. The nucleus exchanges substances with the cytoplasm through nuclear pores.
C. Removal of the nucleus causes the death of the cell.
D. Nuclear division occurs before the division of the cytoplasm.

##C##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMB0030000011|!| ★

Which of the following structures can be found in a plant cell but ***not*** in a prokaryotic cell?

A. ribosomes
B. cell membrane
C. DNA
D. chloroplasts

##D##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMB0030000012|!| ★ **Cross topic**

Which of the following types of cell would contain the greatest relative number of mitochondria?

A. guard cell
B. mesophyll cell
C. root hair cell
D. xylem cell

##C##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

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|!|BCEMB0030000013|!| ★

Which of the following is ***not*** a tissue?

A. skin
B. muscle
C. nerve
D. blood

##A##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMC0030000014|!|

A bacterial cell differs from a mesophyll cell by

(1) having no cell wall.
(2) having no cell membrane.
(3) having no nuclear membrane.
(4) being 100 times smaller.

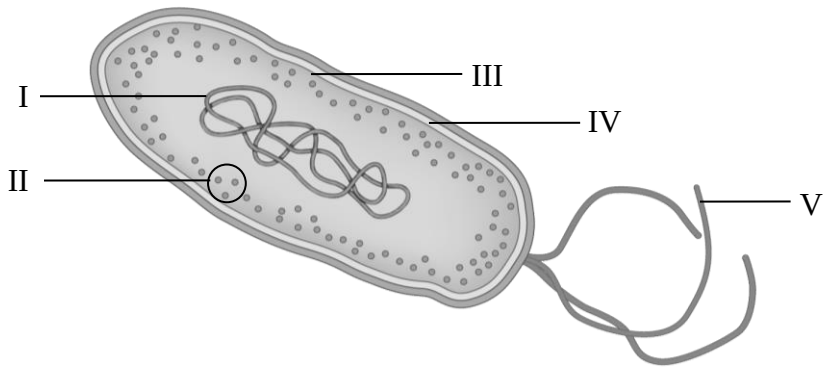
A. (1) and (2) only
B. (2) and (3) only
C. (3) and (4) only
D. (1) and (4) only

##C##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMC0030000015|!|

The diagram below shows the structure of a prokaryotic cell.




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What is the function of structure II?

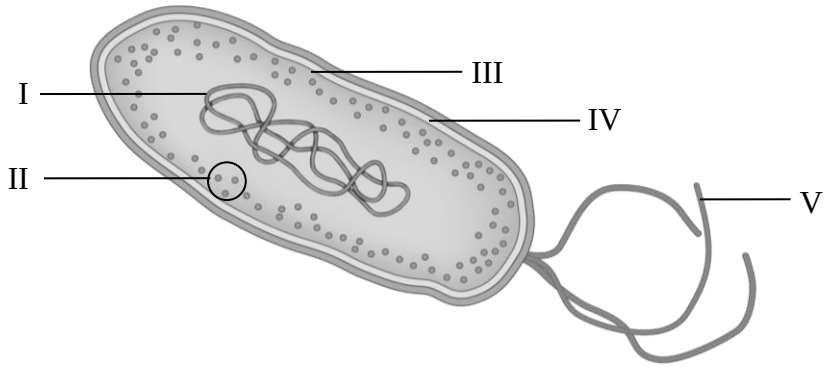
A. energy production
B. storage of nutrients
C. protein synthesis
D. storage of genetic material

##C##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMC0030000016|!|

The diagram below shows the structure of a prokaryotic cell.




Which structures are found in prokaryotic cells and eukaryotic cells?

A. I and II only
B. II and III only
C. I and IV only
D. IV and V only

##B##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

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|!|BCEMB0030000017|!| ★

The following table shows the reagents that can be used to test for different biological molecules.

| Reagents | Biological molecule tested |
|---|---|
| P | lipid |
| Q | starch |
| R | protein |
| S | nucleic acid |

Which of the following reagents will show a positive result if some isolated ribosomes are tested using the above reagents?

A. reagent R only
B. reagents P and Q only
C. reagents R and S only
D. reagents P, R and S only

##C##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMC0030000019|!|

Which of the following substances are synthesized at the ribosomes?

A. steroids
B. polypeptides
C. RNA
D. starch

##B##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMB0030000020|!| ★

Which of the following comparisons between prokaryotic cells and eukaryotic cells is ***incorrect***?

| | ***Prokaryotic cells*** | ***Eukaryotic cells*** |
|---|---|---|
| A. | smaller in size | larger in size |
| B. | example: bacterial cell | example: animal cell |
| C. | do not possess RNA | possess RNA |
| D. | without mitochondria | with mitochondria |

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##C##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000021|!|

Mitochondria are the main sites of

A. respiration.
B. photosynthesis.
C. protein synthesis.
D. lipid synthesis.

##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000022|!|

Which of the following organelles are surrounded by a double membrane?

(1) chloroplast
(2) mitochondrion
(3) nucleus

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000023|!|

What happens when the iris diaphragm of the light microscope is fully opened?

A. The image becomes dimmer.
B. The image becomes brighter.
C. The image becomes sharper.
D. The image becomes more blurred.

##B##

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| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000024|!|
Which of the following is the correct sequence of using a light microscope for observation at low power magnification?

(1) slowly turn the coarse adjustment knob to raise the body tube until the image is clear
(2) lower the body tube by turning the coarse adjustment knob until the objective is just above the slide
(3) adjust the condenser and the iris diaphragm until the light is sufficient and even
(4) turn the fine adjustment knob to make the image sharp

A. (2), (1), (4), (3)
B. (2), (3), (1), (4)
C. (3), (2), (1), (4)
D. (4), (1), (3), (2)

##C##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000025|!|
What is the image formed when the letter ‘q’ is observed with a microscope?

A. b
B. d
C. p
D. q

##A##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

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|!|BCEMC0030000026|!|

Which of the following is the function of the smooth endoplasmic reticulum?

A. synthesis of nucleic acids
B. transport of lipids
C. synthesis of proteins
D. breakdown of glucose

##B##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000027|!|

The following shows the different levels of organization in organisms:

(1) cells
(2) organs
(3) systems
(4) tissues
(5) organelles

Which of the following shows the correct order of the levels from simple to complex in organisms?

A. (1), (4), (2), (3), (5)
B. (3), (2), (4), (1), (5)
C. (5), (1), (4), (2), (3)
D. (5), (2), (1), (4), (3)

##C##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000028|!|

Which of the following is ***not*** a eukaryotic cell?

A. *Escherichia coli* cell
B. *Amoeba* cell
C. human corneal cell
D. onion epidermal cell

##A##

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| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000029|!|

Which of the following statements about the nucleus is ***incorrect***?

A. It is surrounded by a single membrane.

B. It contains the genetic material DNA.

C. It is absent in mature red blood cells.

D. It can be found in eukaryotic cells.

##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000030|!|

Which of the following comparisons between high power magnification and low power magnification is ***incorrect***?

| | | ***High power magnification*** | ***Low power magnification*** |
|---|---|---|---|
| A. | Degree of magnification | larger | smaller |
| B. | Details of the image observed | more | fewer |
| C. | Area of specimen observed | smaller | larger |
| D. | Brightness of image | brighter | dimmer |

##D##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000031|!|

Which of the following cell structures is/are found in a mesophyll cell?

(1) mitochondrion

(2) cell wall

(3) endoplasmic reticulum

A. (2) only

B. (1) and (2) only

C. (2) and (3) only

D. (1), (2) and (3)

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##D##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMB0030000032|!| ★

Which of the following statements about the cell wall is ***incorrect***?

A. It is fully permeable to water.
B. It is made up of cellulose in plant cells.
C. It is absent in fungal cells.
D. It is not made up of cellulose in bacterial cells.

##C##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMB0030000033|!| ★

Which of the following structures is/are absent in animal cells but present in some prokaryotic cells?

(1) chloroplast
(2) cell wall
(3) coiled DNA loop

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMB0030000034|!| ★

Which of the following comparisons between a sperm and a mesophyll cell is ***incorrect***?

| | | ***Sperm*** | ***Mesophyll cell*** |
|---|---|---|---|
| A. | Cell wall | absent | present |
| B. | Ribosome | present | absent |
| C. | Starch granule | absent | present |
| D. | Flagellum | present | absent |

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##B##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMC0030000035|!|

Which of the following is ***not*** a type of plant tissue?

A. epidermal tissue
B. epithelial tissue
C. mesophyll tissue
D. vascular tissue

##B##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000036|!|

Which of the following are the reasons why the specimen in a temporary mount should be as thin as possible?

(1) Fewer layers of cells are observed and this makes the observation of cells easier.
(2) The cover slip can lie flat on the slide.
(3) More light can pass through the specimen, so that the image is bright.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

| Skill assessed | Mark |
|---|---|
| Practical skills | 1 |

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|!|BCEMC0030000037|!|

What is the main function of methylene blue solution in the preparation of a temporary mount of ox corneal cells?

A. to preserve the cells permanently
B. to preserve the colours of the cells
C. to kill the cells
D. to make transparent structures easier to be observed

##D##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000038|!|

The plant cell wall is more rigid than the cell membrane because it contains

A. proteins.
B. phospholipids.
C. cellulose.
D. starch.

##C##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000039|!|

Which of the following is/are the common feature(s) of cell membranes and cell walls?

(1) transparent
(2) flexible
(3) freely permeable to water soluble substances

A. (1) only
B. (3) only
C. (1) and (3) only
D. (1), (2) and (3)

##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

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|!|BCEMC0030000040|!|

Which of the following statements about the nucleus is ***incorrect***?

A. It controls all activities of the cell.
B. It is present in prokaryotic cells.
C. It is the site of mRNA synthesis.
D. It is the site of making ribosomes.

##B##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000041|!|

Which of the following statements about the cell membrane is ***incorrect***?

A. It can be found in bacterial cells.
B. It is made up of phospholipid.
C. It is transparent and inelastic.
D. It is differentially permeable.

##C##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000042|!|

Which of the following cell structures are present in both human white blood cells and plant root cells?

(1) nucleus
(2) mitochondria
(3) ribosomes

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

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|!|BCEMC0030000043|!|

All living cells have

(1) mitochondria.
(2) a cell membrane.
(3) a nucleus.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000044|!|

Which of the following cell structures is/are present in a mature human red blood cell?

(1) nucleus
(2) cytoplasm
(3) mitochondrion

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMB0030000045|!| ★

Which of the following human cells has the lowest density of mitochondria?

A. sperm
B. muscle cell
C. red blood cell
D. liver cell

##C##

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| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMB0030000046|!| ★

Bacterial cells, mesophyll cells and white blood cells all contain

A. a cell wall.
B. a nucleus.
C. ribosomes.
D. rough endoplasmic reticulum.

##C##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMC0030000047|!|

The fruit of a tree can be described as

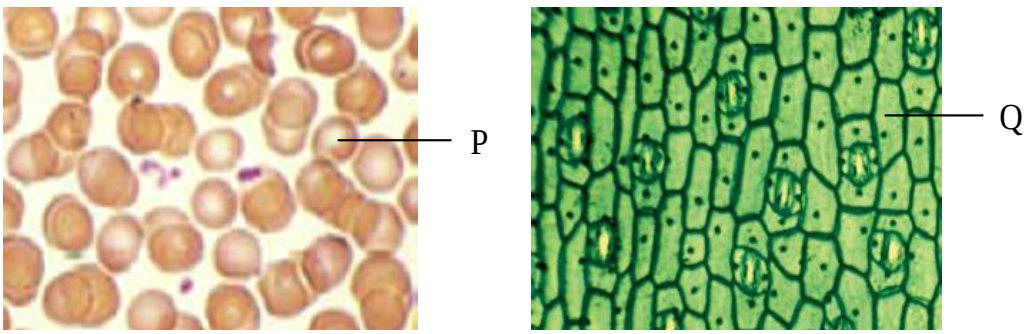
A. a tissue.
B. an organ.
C. a system.
D. an organism.

##B##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

|!|BCEMB0030000048|!| ★

The following photomicrographs show two kinds of cells, P and Q. Cell P is found in humans and cell Q is found in the leaves.




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Which of the following comparisons between cells P and Q is ***incorrect***?

| | ***Cell P*** | ***Cell Q*** |
|---|---|---|
| A. | has no nucleus | has a nucleus |
| B. | has no chloroplasts | has chloroplasts |
| C. | has no cell wall | has a cell wall |
| D. | contains haemoglobin | contains no haemoglobin |

□

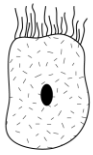
##B##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMB0030000049|!| ★ **Cross topic**

The following diagrams show four types of human cells (not drawn to the same scale):

P



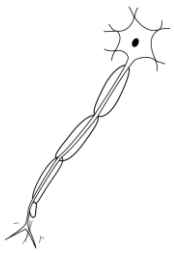
Q



R



S



Which of the following features is ***not*** common to all the cells shown above?

A. mitochondria

B. nuclear membrane

C. diploid number of chromosomes

D. cell membrane that is composed of phospholipid bilayer

□

##C##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

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|!|BCEMC0030000051!|

Which of the following cell structures is/are present in all prokaryotic cells?

(1) nucleus
(2) flagellum
(3) cell membrane

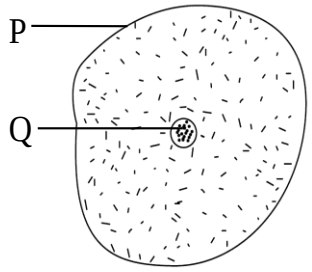
A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

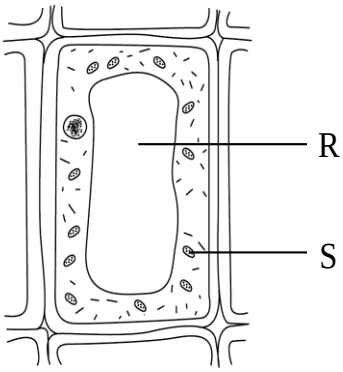
##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMB0030000052|!| ★

The diagrams below show an animal cell and a plant cell.




Which of the following combinations correctly matches the name and function of the labelled structures?

| | ***Structure*** | ***Name*** | ***Function*** |
|---|---|---|---|
| A. | P | cell membrane | prevents the cell from bursting when excess water enters the cell |
| B. | Q | nucleus | supplies energy to the cell |
| C. | R | vacuole | provides support to the plant when it is full |
| D. | S | ribosome | involves in protein synthesis |

##C##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMC0030000053|!|

Which of the following organelles is ***not*** surrounded by a membrane?

A. nucleus
B. ribosome
C. vacuole
D. mitochondrion

##B##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000054|!|

Which of the following is/are the location(s) where ribosomes are found in a eukaryotic cell?

(1) in the cytoplasm
(2) on smooth endoplasmic reticulum
(3) in mitochondria

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000055|!| **Cross topic**

When the vacuole of a plant cell is fully filled with water,

A. the cell will become turgid.
B. the cell will burst.
C. the vacuole will burst.
D. water will leave the cell by osmosis.

##A##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

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|!|BCEMC0030000056|!|

Which of the following statements about the modern cell theory is/are correct?

(1) Cells are specialized for different functions.
(2) All organisms are made up of one or more cells.
(3) The cell is the basic unit of life.

A. (3) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMC0030000057|!|

A student observes a human blood smear with a light microscope. In order to obtain an image with more details of red blood cells, he should

A. increase the distance between the slide and the objective lens.
B. use an objective lens with a higher magnification.
C. use an eyepiece with a lower magnification.
D. increase the illumination of the slide.

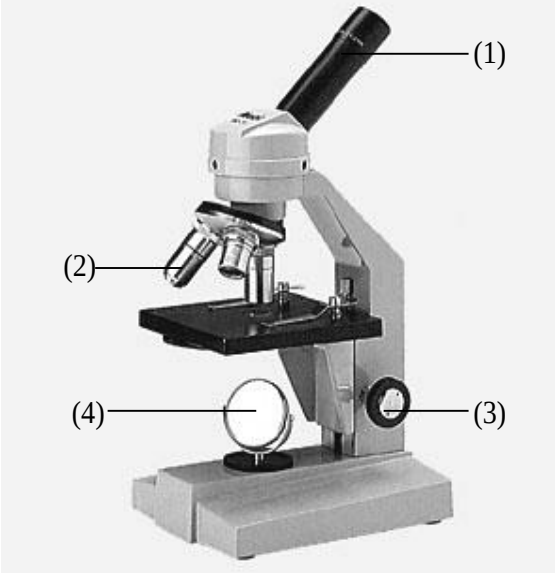
##B##

| Skill assessed | Mark |
|---|---|
| Practical skills | 1 |

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|!|BCEMC0030000058|!|

The following diagram shows a light microscope.




Which part(s) of the light microscope is/are used to adjust the magnification of the light microscope?

A. (2) only
B. (1) and (2) only
C. (3) and (4) only
D. (1), (2) and (3) only

☐

##B##

| Skill assessed | Mark |
|---|---|
| Understanding basic concepts | 1 |

|!|BCEMB0030000060|!| ★

Why should we view the specimen first with low power magnification before switching to high power magnification?

A. It is easier to search for the part we want to observe.
B. The image of the specimen is larger and the specimen can be observed more easily.
C. The image is clearer and the specimen can be located more easily.
D. It poses a lower risk of breaking the slide during focusing.

☐

##A##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

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|!|BCEMC0030000061|!|

Which of the following is ***not*** the function of the cover slip in the preparation of a temporary mount of *Amoeba* cells?

A. to prevent the specimens from drying up
B. to flatten the specimens
C. to hold the specimens in position
D. to prevent the objective from touching the specimens

##C##

| Skill assessed | Mark |
|---|---|
| Practical skills | 1 |

|!|BCEMC0030000063|!|

Which of the following steps of using a light microscope is/are correct?

(1) Open both eyes when observing the specimen with the microscope.
(2) Increase the illumination of the specimen when observing at high power magnification.
(3) Turn the coarse adjustment knob when focusing on the specimen at high power magnification.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

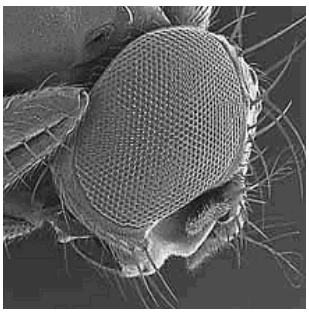
##B##

| Skill assessed | Mark |
|---|---|
| Practical skills | 1 |

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|!|BCEMB0030000064|!| ★

The image below shows the compound eye of a fruit fly.



This image is most likely produced with a

A. transmission electron microscope (TEM).
B. scanning electron microscope (SEM).
C. dissection microscope.
D. compound microscope.

☐

##B##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMB0030000065|!| ★

When you were observing an *Amoeba* with a light microscope, you saw that it moved to the upper right side of the field of view. In which direction did it actually move to, relative to the slide?

A. the upper right side
B. the lower right side
C. the upper left side
D. the lower left side

☐

##D##

| Skill assessed | Mark |
|---|---|
| Applying concepts | 1 |

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|!|BCEMB0030000066|!| ★

Mary observed the onion epidermis with a light microscope. She first focused the specimen and obtained the image as shown in **Figure I**. She then observed the specimen with another magnification and obtained the image as shown in **Figure II**.

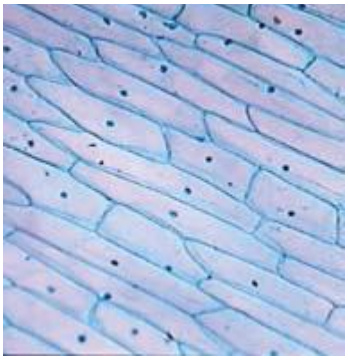


**Figure I**

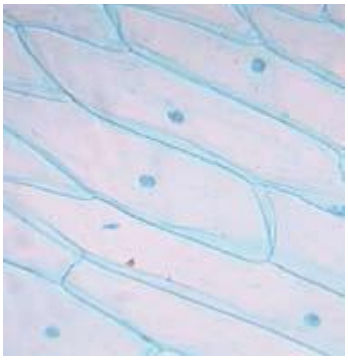


**Figure II**

Which of the following steps is/are necessary for her to obtain the image as shown in **Figure II**?

(1) adjust the iris diaphragm
(2) rotate the nosepiece to select a high power objective
(3) turn the coarse adjustment knob

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

| Skill assessed | Mark |
|---|---|
| Practical skills | 1 |

|!|BCEMB0030000067|!| ★

Which of the following combinations of eyepiece and objective enables you to see the fewest cells in the field of view of the microscope?

| | *Eyepiece* | *Objective* |
|---|---|---|
| A. | 10X | 10X |
| B. | 15X | 15X |
| C. | 10X | 30X |
| D. | 5X | 40X |

##C##

| Skill assessed | Mark |
|---|---|
| Mathematical skills | 1 |

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|!|BCEMB0030000068|!| ★

***Directions:*** The following three questions refer to the table below, which shows the relative amount of sub-cellular structures in four cell types P, Q, R and S.

| Cell type | Relative amount of sub-cellular structure | | |
|---|---|---|---|
| | **Chloroplast** | **Mitochondrion** | **Endoplasmic reticulum** |
| P | – | – | – |
| Q | – | +++ | + |
| R | – | + | +++ |
| S | +++ | +++ | + |

Key: number of '+' indicates the relative amount of sub-cellular structure
'–' indicates the absence of the sub-cellular structure

Cell type P is probably a

A. root epidermal cell.
B. bacterial cell.
C. sperm cell.
D. neurone.

□

##B##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMB0030000069|!| ★

According to the table, which of the following statements best describes cell type R?

A. It is actively participated in protein synthesis.
B. It can carry out photosynthesis.
C. It can carry out both aerobic and anaerobic respiration.
D. It is active in cell division.

□

##A##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

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|!|BCEMB0030000070|!| ★ **Cross topic**

Which of the following cell type would be produced in the testes?

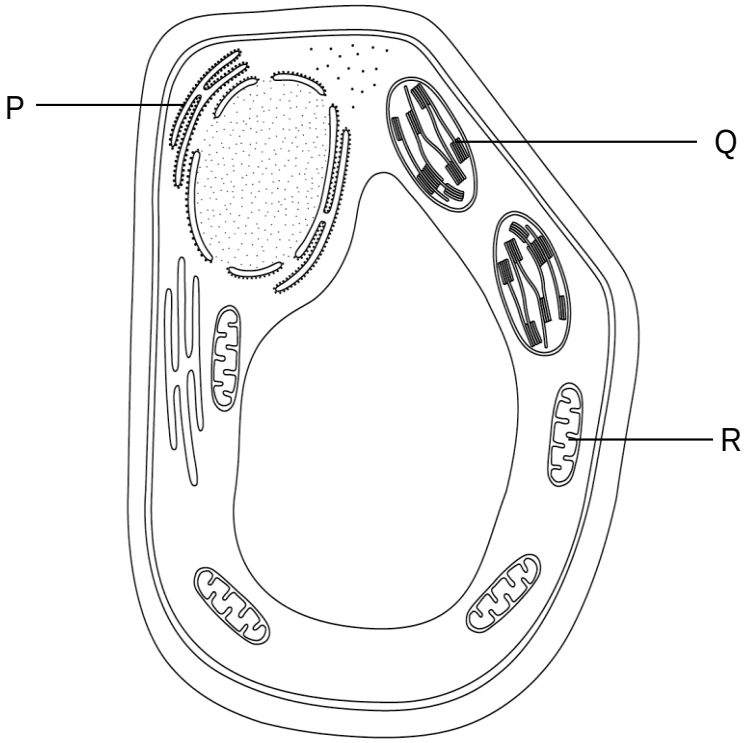
A. P
B. Q
C. R
D. S

##B##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMB0030000071|!| ★

***Directions:*** The following two questions refer to the diagram below which shows a typical plant cell.




Which of the following processes occurs in structure P?

A. aerobic respiration
B. transport of lipid
C. DNA replication
D. protein synthesis

##D##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

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|!|BCEMB0030000072|!| ★

Which of the following comparisons between structures Q and R is ***incorrect***?

| | ***Structure Q*** | ***Structure R*** |
|---|---|---|
| A. | involves in glucose synthesis | involves in glucose breakdown |
| B. | with no ATP synthesis | with ATP synthesis |
| C. | absorbs light energy | does not absorb light energy |
| D. | does not release carbon dioxide | releases carbon dioxide |

##B##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMB0030000073|!| ★

***Directions:*** The following three questions refer to an experiment in which the same prepared slide of *amoebae* was observed in turn with four microscopes with different magnifications. The following table shows the combinations of eyepiece and objective used in different microscopes:

| Microscope | Eyepiece | Objective |
|---|---|---|
| I | 5X | 10X |
| II | 10X | 15X |
| III | 5X | 40X |
| IV | 15X | 10X |

With which microscope could the student observe the largest number of *amoebae*?

A. microscope I
B. microscope II
C. microscope III
D. microscope IV

##A##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

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|!|BCEMB0030000074|!| ★

Under which of the following pairs of microscopes would the *amoebae* appear to move at the same speed?

A. microscopes I and II
B. microscopes II and III
C. microscopes II and IV
D. microscopes III and IV

##C##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMB0030000075|!| ★

Which microscope produces the brightest image with the same light source?

A. microscope I
B. microscope II
C. microscope III
D. microscope IV

##A##

| Skill assessed | Mark |
|---|---|
| Analysing data | 1 |

|!|BCEMB0030000076|!| ★

***Directions:*** The following two questions refer to the following information.

> 'The diameter of the field of view of a microscope at high power magnification (400X) was measured and found to be 0.2 mm.'

When a student used the high power microscope (400X) to observe the epidermis of a leaf, he could find two stomata in the field of view. What is the average density of stomata on the epidermis (correct to the nearest digit)?

A. 16 stomata / $mm^2$
B. 50 stomata / $mm^2$
C. 64 stomata / $mm^2$
D. 200 stomata / $mm^2$

##C##

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| Skill assessed | Mark |
|---|---|
| Mathematical skills | 1 |

|!|BCEMB0030000077|!| ★

The student then uses the high power microscope (400X) to observe a blood smear. It is known that a human red blood cell is approximately 8 μm (0.008 mm) in diameter. If the red blood cells line up, edge to edge, in a row at the diameter of the field of view, how many red blood cells are in the row (correct to the nearest digit)?

A. 160
B. 50
C. 25
D. 13

□

##C##

| Skill assessed | Mark |
|---|---|
| Mathematical skills | 1 |

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